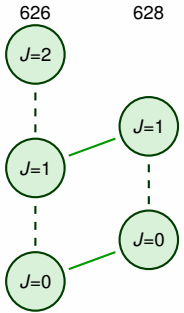
1. Physical Graph Construction

Node = energy state
Features: E, J, parity, TROVE
$d_{\max}$, t_1, t_2, t_3, P, sym., masses



— inter-isotopologue (perturbation)

- - intra-isotopologue (■J=1 ladder)

2. GraphSAGE Architecture

Hidden dim 256; Focal Loss
(γ=2.0); `NeighborLoader`([15,10])

12 features ⊕ 8-d isotopologue embedding

↓

Linear projection → 256

↓

SAGEConv → LayerNorm → GELU → Dropout
+ residual (× 2 layers)

↓

Head: Linear → LN → GELU → Dropout(0.3) → Linear

↓

Softmax → $\hat{p}_c$ per class, per node

3. Hungarian Uniqueness Solver

Bijective assignment within each physical block; $\hat{p}_{\text{assigned}}$ suppressed for conflicted nodes

Cost $C_{ij} = 1 - P_{ij}$, one (isotope, J, parity, P) block

	c_1	c_2	c_3
s_1	0.10	0.70	0.85
s_2	0.15	0.40	0.90
s_3	0.60	0.55	0.20

Naive argmax: $s_1 \to c_1$ *and* $s_2 \to c_1$ conflict (duplicate label)

Hungarian optimum: $s_1 \to c_1$, $s_2 \to c_2$, $s_3 \to c_3$ (unique, min. total cost)